Continuous Map.

Def. A map $f: X \to Y$ is said to be Continuous at $x_0 \in X$ if:
For any open $V \subseteq Y$ with $f(x_0) \in V$, there exists an open $\mathcal{U} \subseteq X$ with $x_0 \in \mathcal{U}$ s.t $f[\mathcal{U}] \subseteq V$

Def. A map $f: X \to Y$ is called Continuous if:
For any open $V \subseteq Y$, $f^{-1}[V] \subseteq X$ is open of X.

Proposition.

$f: X \to Y$ is Continuous if and only if for any $x_0 \in X$, f is continuous at x_0.

Proof. observe that if $V \subseteq Y$ Contains $f(x_0)$, then

$$x_0 \in f^{-1}[f[\{x_0\}]] \subseteq f^{-1}[V]$$

Thus, directly $\Rightarrow$) is proven. Conversely, let $V \subseteq Y$ be an open set.
For any $y \in f^{-1}[V]$, $f(y) \in V$ implies there exists open $\mathcal{U}_y$ containing y s.t

$$f[\mathcal{U}_y] \subseteq V \quad \text{which implies } \mathcal{U}_y \subseteq f^{-1}[f[\mathcal{U}_y]] \subseteq f^{-1}[V]$$

Now, $f^{-1}[V] = \bigcup_{y \in f^{-1}[V]} \mathcal{U}_y$ is open.

Thm. Let $f: X \to Y$ be a map. TFAE:

a) f is continuous.

b) For any element B of basis of Y, $f^{-1}[B] \subseteq X$ is open.

c) For any $A \subseteq X$, $f[\overline{A}] \subseteq \overline{f[A]}$

d) For any $B \subseteq Y$, $\overline{f^{-1}[B]} \subseteq f^{-1}[\overline{B}]$

e) For any closed $C \subseteq Y$, $f^{-1}[C]$ is closed.

Proof. a) $\Rightarrow$ b) is trivial, and b) $\Rightarrow$ a) is given by: if $V = \bigcup B$ is open, then

$$f^{-1}[V] = f^{-1}[\bigcup B] = \bigcup f^{-1}[B], \text{ open.}$$

a) $\Leftrightarrow$ e) is given by $f^{-1}[Y \setminus C] = X \setminus f^{-1}[C]$. But, c) and d) are not obvious.
Suppose f is continuous, and $A \subseteq X$. Let $f(x) \in f[\overline{A}]$ be given. Since $x \in \overline{A}$,

$$\text{for any open } \mathcal{U} \subseteq X \text{ s.t. } x \in \mathcal{U}, \quad \mathcal{U} \cap A \neq \emptyset$$

Let $V \subseteq Y$ be an open s.t. $f(x) \in V$. Since f is continuous, $f^{-1}[V] \subseteq X$ is open, and

$$x \in f^{-1}[f[\{x\}]] \subseteq f^{-1}[V]$$

Now,

$$V \cap f[A] \supseteq f[f^{-1}[V]] \cap f[A] \supseteq f[f^{-1}[V] \cap A] \neq \emptyset$$

Therefore $f(x) \in \overline{f[A]}$. Suppose c) hold, and let $B \subseteq Y$ be given.
Then,

$$f[\overline{f^{-1}[B]}] \subseteq \overline{f[f^{-1}[B]]} \subseteq \overline{B}$$

Therefore

$$\overline{f^{-1}[B]} \subseteq f^{-1}[f[\overline{f^{-1}[B]}]] \subseteq f^{-1}[\overline{B}].$$

Finally, suppose that d) holds. Then, for any closed $C \subseteq Y$,

$$\overline{f^{-1}[C]} \subseteq f^{-1}[\overline{C}] = f^{-1}[C] \text{ and trivially } f^{-1}[C] \subseteq \overline{f^{-1}[C]}.$$

That is, $f^{-1}[C]$ is closed, thus f is continuous.

「Thm. Let $f: X \to Y$ be a continuous map.

If a sequence $\{x_n\}$ in X converges to x_0, then $\{f(x_n)\}$ converges to $f(x_0)$.

Proof By the assumption, given open $x_0 \in \mathcal{U} \subseteq X$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow x_n \in \mathcal{U}$.
Let an open $V \subseteq Y$ satisfying $f(x_0) \in V$ be given. Since f is continuous, $f^{-1}[V]$ is open with

$$x_0 \in f^{-1}[\{f(x_0)\}] \subseteq f^{-1}[V]$$

Now, there exists $N_0 \in \mathbb{N}$ s.t $n \geq N_0 \Rightarrow x_n \in f^{-1}[V]$, which means $f(x_n) \in f[f^{-1}[V]] \subseteq V$ ∎

But,

「Counterexample $x_n \to x$ implies $f(x_n) \to f(x)$ but f is <u>not</u> continuous.
Define a map on co-countable space $X = (\mathbb{R}, \tau_c)$ into discrete $Y = (\{0,1\}, \mathcal{P}(\{0,1\}))$ as:

$$f: X \to Y;\ x \mapsto \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Let $\{x_n\}$ be a sequence in X converges to $x \in X$. Since

$$B = \{x_k \mid x_k \neq x\}$$

is countable subset of X and $x \notin B$, $X \setminus B$ is open neighborhood of x.
Since $x_n \to x$, there exists $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow x_n \in X \setminus B$, which implies $x_n = x$, $\forall n \geq N$.
Now, regardless of $x \in \mathbb{Q}$ or $x \notin \mathbb{Q}$, $f(x_n) \to f(x)$. But f is not continuous, because
$f^{-1}[\{0\}] = \mathbb{Q}$ is not open

「Pasting Lemma.
Let A, B be opens or closeds. s.t $X = A \cup B$.
If $f: A \to Y$ conti, $g: B \to Y$ conti, $f|_{A \cap B} = g|_{A \cap B}$ then

$$f \cup g: X \to Y;\ x \mapsto \begin{cases} f(x) & x \in A \\ g(x) & x \in B \end{cases}.$$

is continuous